

PRÀCTICA:

Analitzador de protocols Wireshark

1.

The image displays two screenshots of the Wireshark network protocol analyzer interface, specifically showing ICMP Echo (ping) traffic. The top screenshot shows a list of captured packets, with packet 706 selected. The bottom screenshot shows the detailed view of packet 707, which is an ICMP Echo (ping) reply.

Top Screenshot: Packet List

No.	Time	Source	Destination	Protocol	Length	Info
352	22.541250182	192.168.0.12	212.166.132.192	ICMP	154	Destination unreachable (Port unreachable)
706	74.891803537	192.168.0.20	8.8.8.8	ICMP	98	Echo (ping) request id=0x09ba, seq=1/256, ttl=64 (reply in 7)
707	74.910661726	8.8.8.8	192.168.0.20	ICMP	98	Echo (ping) reply id=0x09ba, seq=1/256, ttl=118 (request i)
712	75.887321007	192.168.0.20	8.8.8.8	ICMP	98	Echo (ping) request id=0x09ba, seq=2/512, ttl=64 (reply in 7)
715	76.014769518	8.8.8.8	192.168.0.20	ICMP	98	Echo (ping) reply id=0x09ba, seq=2/512, ttl=118 (request i)
716	76.890819007	192.168.0.20	8.8.8.8	ICMP	98	Echo (ping) request id=0x09ba, seq=3/768, ttl=64 (reply in 7)
717	76.914664732	8.8.8.8	192.168.0.20	ICMP	98	Echo (ping) reply id=0x09ba, seq=3/768, ttl=118 (request i)
726	77.894317005	192.168.0.20	8.8.8.8	ICMP	98	Echo (ping) request id=0x09ba, seq=4/1024, ttl=64 (reply in 7)
729	77.952273372	8.8.8.8	192.168.0.20	ICMP	98	Echo (ping) reply id=0x09ba, seq=4/1024, ttl=118 (request i)
770	78.372753235	192.168.0.12	212.166.132.192	ICMP	167	Destination unreachable (Port unreachable)
2072	145.954906883	192.168.0.12	212.166.132.192	ICMP	181	Destination unreachable (Port unreachable)
3694	226.570109251	192.168.0.12	212.166.132.96	ICMP	268	Destination unreachable (Port unreachable)

Bottom Screenshot: Packet Details (Frame 707)

Frame 707: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface 0

- Ethernet II, Src: PCSSystemtec_Ba:35:d2 (08:00:27:8a:35:d2), Dst: Sa
- Internet Protocol Version 4, Src: 192.168.0.20, Dst: 8.8.8.8
- Internet Control Message Protocol
 - Type: Echo (ping) request (8)
 - Code: 0
 - Checksum: 0x51d0 [correct]
 - [Checksum Status: Good]
 - Identifier (BE): 2490 (0x09ba)
 - Identifier (LE): 47625 (0xba09)
 - Sequence Number (BE): 1 (0x0001)
 - Sequence Number (LE): 256 (0x0100)
 - [Response frame: 707]
 - ICMP Data: 9dfd136a0000000001d3a0f0000000000101112131415161718191a:

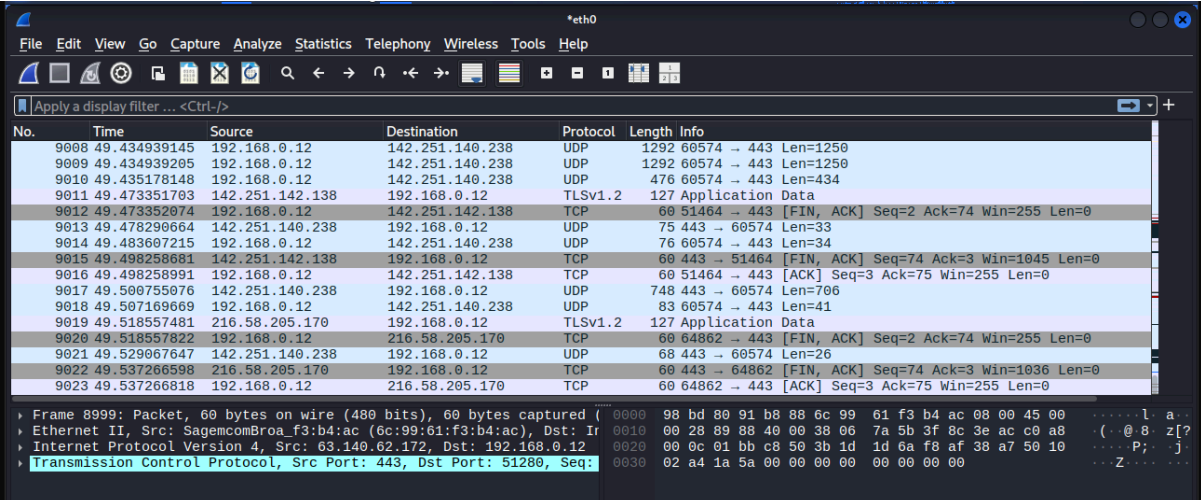
Bottom Screenshot: Packet Details (Frame 707) - Reply

Frame 707: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface 0

- Ethernet II, Src: SagemcomBroa_f3:b4:ac (6c:99:61:f3:b4:ac), Dst: PC
- Internet Protocol Version 4, Src: 8.8.8.8, Dst: 192.168.0.20
- Internet Control Message Protocol
 - Type: Echo (ping) reply (0)
 - Code: 0
 - Checksum: 0x59d0 [correct]
 - [Checksum Status: Good]
 - Identifier (BE): 2490 (0x09ba)
 - Identifier (LE): 47625 (0xba09)
 - Sequence Number (BE): 1 (0x0001)
 - Sequence Number (LE): 256 (0x0100)
 - [Request frame: 706]
 - [Response time: 18.858 ms]
 - ICMP Data: 9dfd136a0000000001d3a0f0000000000101112131415161718191a:

Al paquet de petició d'eco (Echo ping request), si despleguem el protocol Internet Control Message Protocol, podem observar que utilitza el Type 8. Per contra, al paquet de resposta (Echo ping reply), el camp Type és 0. Això ens permet diferenciar visualment quin equip demana la connexió i quin respon.

2.

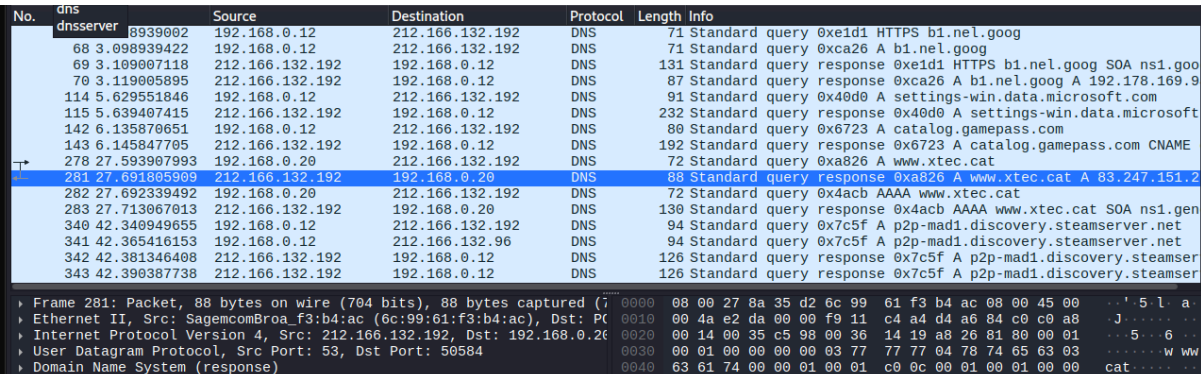


No.	Time	Source	Destination	Protocol	Length	Info
9008	49.434939145	192.168.0.12	142.251.140.238	UDP	1292	60574 → 443 Len=1250
9009	49.434939205	192.168.0.12	142.251.140.238	UDP	1292	60574 → 443 Len=1250
9010	49.435178148	192.168.0.12	142.251.140.238	UDP	476	60574 → 443 Len=434
9011	49.473351703	142.251.142.138	192.168.0.12	TLSv1.2	127	Application Data
9012	49.473352074	192.168.0.12	142.251.142.138	TCP	60	51464 → 443 [FIN, ACK] Seq=2 Ack=74 Win=255 Len=0
9013	49.478290664	142.251.140.238	192.168.0.12	UDP	75	443 → 60574 Len=33
9014	49.483607215	192.168.0.12	142.251.140.238	UDP	76	60574 → 443 Len=34
9015	49.498258681	142.251.142.138	192.168.0.12	TCP	60	443 → 51464 [FIN, ACK] Seq=74 Ack=3 Win=1045 Len=0
9016	49.498258991	192.168.0.12	142.251.142.138	TCP	60	51464 → 443 [ACK] Seq=3 Ack=75 Win=255 Len=0
9017	49.500755076	142.251.140.238	192.168.0.12	UDP	748	443 → 60574 Len=706
9018	49.507169669	192.168.0.12	142.251.140.238	UDP	83	60574 → 443 Len=41
9019	49.518557481	216.58.205.170	192.168.0.12	TLSv1.2	127	Application Data
9020	49.518557822	192.168.0.12	216.58.205.170	TCP	60	64862 → 443 [FIN, ACK] Seq=2 Ack=74 Win=255 Len=0
9021	49.529067647	142.251.140.238	192.168.0.12	UDP	68	443 → 60574 Len=26
9022	49.537266598	216.58.205.170	192.168.0.12	TCP	60	443 → 64862 [FIN, ACK] Seq=74 Ack=3 Win=1036 Len=0
9023	49.537266818	192.168.0.12	216.58.205.170	TCP	60	64862 → 443 [ACK] Seq=3 Ack=75 Win=255 Len=0

Frame 8999: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0
Ethernet II, Src: SagemcomBroadcom_8c:99:61:f3:b4:ac, Dst: Intel_08:00:27:8a:35:d2, Len: 60
Internet Protocol Version 4, Src: 63.140.02.172, Dst: 192.168.0.12
Transmission Control Protocol, Src Port: 443, Dst Port: 51280, Seq: 2

Per a aquest apartat, he activat el mode promiscu a la targeta de xarxa de la màquina virtual. La meua màquina Kali té la IP 192.168.0.20, però com es veu a la captura adjunta, Wireshark està aconseguint interceptar el trànsit de la IP 192.168.0.12, que correspon a la meua màquina física. A la imatge es pot observar el trànsit generat en navegar per internet des de fora de la màquina virtual, capturant peticions a servidors externs mitjançant els protocols TLSv1.2, TCP i UDP a través del port 443 (HTTPS).

3 i 4.



No.	Time	Source	Destination	Protocol	Length	Info
68	3.098939422	192.168.0.12	212.166.132.192	DNS	71	Standard query 0xe1d1 HTTPS b1.nel.goog
69	3.109007118	212.166.132.192	192.168.0.12	DNS	71	Standard query response 0xca26 A b1.nel.goog
70	3.119005895	212.166.132.192	192.168.0.12	DNS	131	Standard query response 0xca26 A b1.nel.goog S0A ns1.google.com
114	5.629551846	192.168.0.12	212.166.132.192	DNS	87	Standard query 0x40d0 A settings-win.data.microsoft.com
115	5.639407415	212.166.132.192	192.168.0.12	DNS	91	Standard query response 0x40d0 A settings-win.data.microsoft.com
142	6.135870651	192.168.0.12	212.166.132.192	DNS	232	Standard query response 0x40d0 A settings-win.data.microsoft.com
143	6.145847705	212.166.132.192	192.168.0.12	DNS	80	Standard query 0x6723 A catalog.gamepass.com
143	6.145847705	212.166.132.192	192.168.0.12	DNS	192	Standard query response 0x6723 A catalog.gamepass.com CNAME
278	27.593907993	192.168.0.20	212.166.132.192	DNS	72	Standard query 0xa826 A www.xtec.cat
281	27.691805909	212.166.132.192	192.168.0.20	DNS	88	Standard query response 0xa826 A www.xtec.cat A 83.247.151.2
282	27.692339492	192.168.0.20	212.166.132.192	DNS	72	Standard query 0x4acb AAAA www.xtec.cat
283	27.713067013	212.166.132.192	192.168.0.20	DNS	130	Standard query response 0x4acb AAAA www.xtec.cat S0A ns1.google.com
340	42.340949655	192.168.0.12	212.166.132.192	DNS	94	Standard query 0x7c5f A p2p-mad1.discovery.steamserver.net
341	42.365416153	192.168.0.12	212.166.132.96	DNS	94	Standard query 0x7c5f A p2p-mad1.discovery.steamserver.net
342	42.381346408	212.166.132.192	192.168.0.12	DNS	126	Standard query response 0x7c5f A p2p-mad1.discovery.steamserver.net
343	42.396387738	212.166.132.192	192.168.0.12	DNS	126	Standard query response 0x7c5f A p2p-mad1.discovery.steamserver.net

Frame 281: Packet, 88 bytes on wire (704 bits), 88 bytes captured (704 bits) on interface eth0
Ethernet II, Src: SagemcomBroadcom_8c:99:61:f3:b4:ac, Dst: Intel_08:00:27:8a:35:d2, Len: 88
Internet Protocol Version 4, Src: 212.166.132.192, Dst: 192.168.0.20
User Datagram Protocol, Src Port: 53, Dst Port: 50584
Domain Name System (response)

A la captura adjunta podem veure la petició DNS (Standard query). El que està passant aquí és que el meu ordinador client li està preguntant al servidor DNS quina és l'adreça IP que correspon al domini www.xtec.cat. Com que els ordinadors es comuniquen mitjançant adreces IP i no amb noms, aquesta traducció és el primer pas necessari abans de poder carregar la pàgina web.

dnsserver	6939002	192.168.0.12	212.166.132.192	DNS	71 Standard query 0xe1d1 HTTPS b1.nel.goog
	68 3.098939422	192.168.0.12	212.166.132.192	DNS	71 Standard query 0xca26 A b1.nel.goog
	69 3.109007118	212.166.132.192	192.168.0.12	DNS	131 Standard query response 0xe1d1 HTTPS b1.nel.goog SOA ns1.goo
	70 3.119005895	212.166.132.192	192.168.0.12	DNS	87 Standard query response 0xca26 A b1.nel.goog A 192.178.169.9
	114 5.629551846	192.168.0.12	212.166.132.192	DNS	91 Standard query 0x40d0 A settings-win.data.microsoft.com
	115 5.639407415	212.166.132.192	192.168.0.12	DNS	232 Standard query response 0x40d0 A settings-win.data.microsoft.com
	142 6.135870651	192.168.0.12	212.166.132.192	DNS	80 Standard query 0x6723 A catalog.gamepass.com
	143 6.145847705	212.166.132.192	192.168.0.12	DNS	192 Standard query response 0x6723 A catalog.gamepass.com CNAME
	278 27.593907993	192.168.0.20	212.166.132.192	DNS	72 Standard query 0xa826 A www.xtec.cat
	281 27.691805909	212.166.132.192	192.168.0.20	DNS	88 Standard query response 0xa826 A www.xtec.cat A 83.247.151.2
	282 27.692339492	192.168.0.20	212.166.132.192	DNS	72 Standard query 0x4acb AAAA www.xtec.cat
	283 27.713067013	212.166.132.192	192.168.0.20	DNS	130 Standard query response 0x4acb AAAA www.xtec.cat SOA ns1.gen
	340 42.340949655	192.168.0.12	212.166.132.192	DNS	94 Standard query 0x7c5f A p2p-mad1.discovery.steamserver.net
	341 42.365416153	192.168.0.12	212.166.132.96	DNS	94 Standard query 0x7c5f A p2p-mad1.discovery.steamserver.net
	342 42.381346408	212.166.132.192	192.168.0.12	DNS	126 Standard query response 0x7c5f A p2p-mad1.discovery.steamser
	343 42.390387738	212.166.132.192	192.168.0.12	DNS	126 Standard query response 0x7c5f A p2p-mad1.discovery.steamser

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Frame 282: Packet, 72 bytes on wire (576 bits), 72 bytes captured (576 bits) on interface 0
Ethernet II, Src: PCSSystemtec_8a:35:d2 (08:00:27:8a:35:d2), Dst: SagemcomBroa_f3:b4:ac (6c:99:61:f3:b4:ac)
Internet Protocol Version 4, Src: 192.168.0.20, Dst: 212.166.132.192
User Datagram Protocol, Src Port: 59863, Dst Port: 53
Domain Name System (query)
0000  6c 99 61 f3 b4 ac 08 00 27 8a 35 d2 08 00 45 00  l a ..... 5
0010  00 3a e2 81 00 00 40 11 7e 0e c0 a8 00 14 d4 a6  : ... @ ...
0020  84 c0 e9 d7 00 35 00 26 1a 5b 4a cb 01 00 00 01  : ... 5 & [ J
0030  00 00 00 00 00 00 03 77 77 77 04 78 74 65 63 03  : ..... w ww
0040  63 61 74 00 00 1c 00 01                          cat .....

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En aquesta segona captura es mostra la resposta del servidor DNS (Standard query response). Després de rebre la nostra petició, el servidor busca a la seva base de dades i ens contesta donant-nos l'adreça IP exacta on es troba allotjada la web de www.xtec.cat. Aquesta IP la podem veure clarament dins de la secció 'Answers' (Respostes) del paquet capturat. Amb aquesta informació, el nostre navegador ja sap a quina IP s'ha de connectar.

5.

No.	Time	Source	Destination	Protocol	Length	Info
279	27.690635488	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.20? Tell 192.168.0.1
280	27.690649908	PCSSystemtec_8a:35:...	SagemcomBroa_f3:b4:...	ARP	42	192.168.0.20 is at 08:00:27:8a:35:d2
289	31.571806784	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.12? Tell 192.168.0.1
290	31.571807084	Intel_91:b8:88	SagemcomBroa_f3:b4:...	ARP	60	192.168.0.12 is at 98:bd:80:91:b8:88
294	32.646563827	PCSSystemtec_8a:35:...	SagemcomBroa_f3:b4:...	ARP	42	Who has 192.168.0.1? Tell 192.168.0.20
295	32.648443253	SagemcomBroa_f3:b4:...	PCSSystemtec_8a:35:...	ARP	60	192.168.0.1 is at 6c:99:61:f3:b4:ac
297	33.274786570	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.12? Tell 192.168.0.1
298	33.274787091	Intel_91:b8:88	SagemcomBroa_f3:b4:...	ARP	60	192.168.0.12 is at 98:bd:80:91:b8:88
7891	579.501061626	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.10? Tell 192.168.0.1
7892	579.501062107	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.11? Tell 192.168.0.1
7893	579.570040461	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.12? Tell 192.168.0.1
7894	579.570040952	Intel_91:b8:88	SagemcomBroa_f3:b4:...	ARP	60	192.168.0.12 is at 98:bd:80:91:b8:88
7895	579.620833353	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.13? Tell 192.168.0.1
7904	581.216169580	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.13? Tell 192.168.0.1
7905	581.321622419	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.18? Tell 192.168.0.1
7914	583.391970686	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.100.1? Tell 192.168.0.1

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Frame 279: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
Ethernet II, Src: SagemcomBroa_f3:b4:ac (6c:99:61:f3:b4:ac), Dst: Broadcast
Address Resolution Protocol (request)
0000  ff ff ff ff ff ff 6c 99 61 f3 b4 ac 08 00 00 01  : ..... l a
0010  08 00 06 04 00 01 0c 99 61 f3 b4 ac c0 a8 00 01  : ..... l a
0020  00 00 00 00 00 00 c0 a8 00 14 00 00 00 00 00  : .....
0030  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  : .....

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A la captura veiem com funciona el protocol ARP. Bàsicament, el meu ordinador pregunta en veu alta qui té la IP del router (192.168.0.1). A la línia 295, el router li contesta. > Dins d'aquesta resposta podem veure clarament que l'adreça MAC del router és 6c:99:61:f3:b4:ac. A més, Wireshark reconeix automàticament aquesta adreça i ens indica que el fabricant de la targeta és Sagemcom (Sagemcom Broadband).

6.

No.	Time	Source	Destination	Protocol	Length	Info
279	27.698635488	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.20? Tell 192.168.0.1
280	27.698649908	PCSSystemtec_8a:35:...	SagemcomBroa_f3:b4:...	ARP	42	192.168.0.20 is at 08:00:27:8a:35:d2
289	31.571896784	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.12? Tell 192.168.0.1
290	31.571897084	Intel_91:b8:88	SagemcomBroa_f3:b4:...	ARP	60	192.168.0.12 is at 98:bd:80:91:b8:88
294	32.646563827	PCSSystemtec_8a:35:...	SagemcomBroa_f3:b4:...	ARP	42	Who has 192.168.0.1? Tell 192.168.0.20
295	32.648443253	SagemcomBroa_f3:b4:...	PCSSystemtec_8a:35:...	ARP	60	192.168.0.1 is at 6c:99:61:f3:b4:ac
297	33.274786570	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.12? Tell 192.168.0.1
298	33.274787091	Intel_91:b8:88	SagemcomBroa_f3:b4:...	ARP	60	192.168.0.12 is at 98:bd:80:91:b8:88
7891	579.501061626	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.10? Tell 192.168.0.1
7892	579.501062107	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.11? Tell 192.168.0.1
7893	579.570040461	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.12? Tell 192.168.0.1
7894	579.570040952	Intel_91:b8:88	SagemcomBroa_f3:b4:...	ARP	60	192.168.0.12 is at 98:bd:80:91:b8:88
7895	579.620833353	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.13? Tell 192.168.0.1
7904	581.216169580	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.13? Tell 192.168.0.1
7905	581.321622419	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.0.18? Tell 192.168.0.1
7914	583.391970686	SagemcomBroa_f3:b4:...	Broadcast	ARP	60	Who has 192.168.100.1? Tell 192.168.0.1

▶ Frame 279: Packet, 60 bytes on wire (480 bits), 60 bytes captured (2...
 ▶ Ethernet II, Src: SagemcomBroa_f3:b4:ac (6c:99:61:f3:b4:ac), Dst: Br...
 ▶ Address Resolution Protocol (request)

"Per resoldre aquest punt, he aplicat el filtre arp.src.proto_ipv4 == 192.168.1.1 a la captura1.pcapng per veure només l'activitat ARP d'aquesta IP. Al paquet número 19, que és una resposta ARP, s'indica clarament ('192.168.1.1 is at...') que l'adreça MAC d'aquest equip és d4:76:ea:0f:fd:58."

7

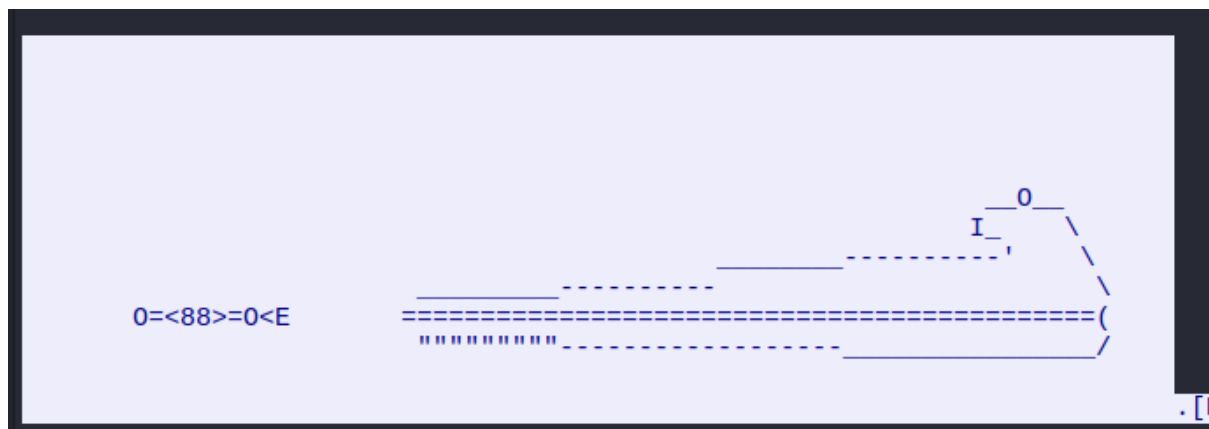
2964	60.870687662	195.135.221.132	192.168.1.147	FTP	85	Response: 220 "Welcome to ftp.suse.com"
3030	65.204216960	192.168.1.147	195.135.221.132	FTP	70	Request: USER anonymous
3032	65.292815488	195.135.221.132	192.168.1.147	FTP	88	Response: 331 Please specify the password.
3050	74.140324720	192.168.1.147	195.135.221.132	FTP	67	Request: PASS contra
3051	74.231585997	195.135.221.132	192.168.1.147	FTP	77	Response: 230 Login successful.
3053	74.231796717	192.168.1.147	195.135.221.132	FTP	60	Request: SYST
3054	74.318835494	195.135.221.132	192.168.1.147	FTP	73	Response: 215 UNIX Type: L8
3057	80.812747274	192.168.1.147	195.135.221.132	FTP	82	Request: PORT 192,168,1,147,137,235
3058	80.902262401	195.135.221.132	192.168.1.147	FTP	105	Response: 200 PORT command successful. Consider using PA

▶ Frame 3030: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on inte...
 ▶ Ethernet II, Src: ASUSTekCOMPU 62:5f:7c (14:da:e9:62:5f:7c), Dst: zte_0f:fd:58 (d4:76...
 ▶ Internet Protocol Version 4, Src: 192.168.1.147, Dst: 195.135.221.132
 ▶ Transmission Control Protocol, Src Port: 59036, Dst Port: 21, Seq: 1, Ack: 32, Len: 1
 ▶ File Transfer Protocol (FTP)
 * USER anonymous\r\n
 Request command: USER
 Request arg: anonymous
 [Current working directory:]

Al filtrar la captura pel protocol ftp, podem comprovar com aquest protocol envia les credencials en text pla. Al paquet 3050 veiem la petició PASS que revela que el password de l'usuari és 'contra'. Més endavant, veiem la comanda RETR, que s'utilitza per descarregar fitxers, la qual ens indica que el fitxer que es descarrega es diu README.txt.

No.	Time	Source	Destination	Protocol	Length	Info
3072	81.174770337	195.135.221.132	192.168.1.147	FTP	78	Response: 226 Directory send OK.
3079	89.795717434	192.168.1.147	195.135.221.132	FTP	63	Request: CWD pub
3081	89.886715104	195.135.221.132	192.168.1.147	FTP	91	Response: 250 Directory successfully changed.
3084	93.172338455	192.168.1.147	195.135.221.132	FTP	81	Request: PORT 192,168,1,147,199,90
3085	93.258954503	195.135.221.132	192.168.1.147	FTP	105	Response: 200 PORT command successful. Consider using PA
3087	93.259169595	192.168.1.147	195.135.221.132	FTP	60	Request: LIST
3095	93.435694620	195.135.221.132	192.168.1.147	FTP	93	Response: 150 Here comes the directory listing.
3099	93.524357087	195.135.221.132	192.168.1.147	FTP	78	Response: 226 Directory send OK.
3244	120.412798389	192.168.1.147	195.135.221.132	FTP	62	Request: TYPE I
3246	120.501895325	195.135.221.132	192.168.1.147	FTP	85	Response: 200 Switching to Binary mode.
3248	120.502113747	192.168.1.147	195.135.221.132	FTP	81	Request: PORT 192,168,1,147,135,85
3249	120.595129311	195.135.221.132	192.168.1.147	FTP	105	Response: 200 PORT command successful. Consider using PA
3250	120.595395003	192.168.1.147	195.135.221.132	FTP	71	Request: RETR README.txt
3262	120.778810403	195.135.221.132	192.168.1.147	FTP	124	Response: 150 Opening BINARY mode data connection for RE
3266	120.869096185	195.135.221.132	192.168.1.147	FTP	78	Response: 226 Transfer complete.
3271	128.900684486	192.168.1.147	195.135.221.132	FTP	60	Request: QUIT

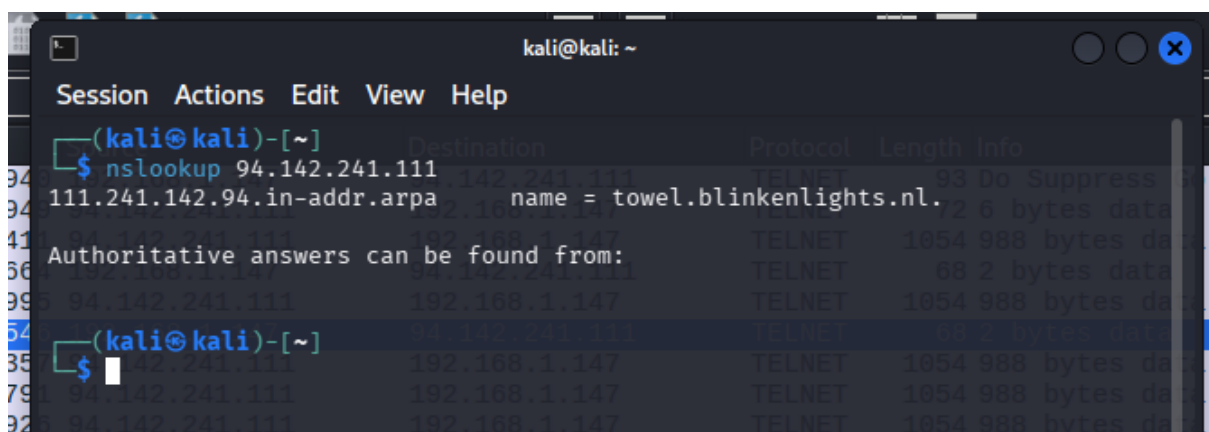
8.



Sí que es pot veure perfectament el que veia l'usuari en connectar-se, ja que el protocol Telnet no xifra la informació i envia les dades en text pla. El que s'observa és una animació de la pel·lícula Star Wars feta amb art ASCII. Els caràcters que componen la nau espacial petita són exactament aquests: 0=<88>=0<E

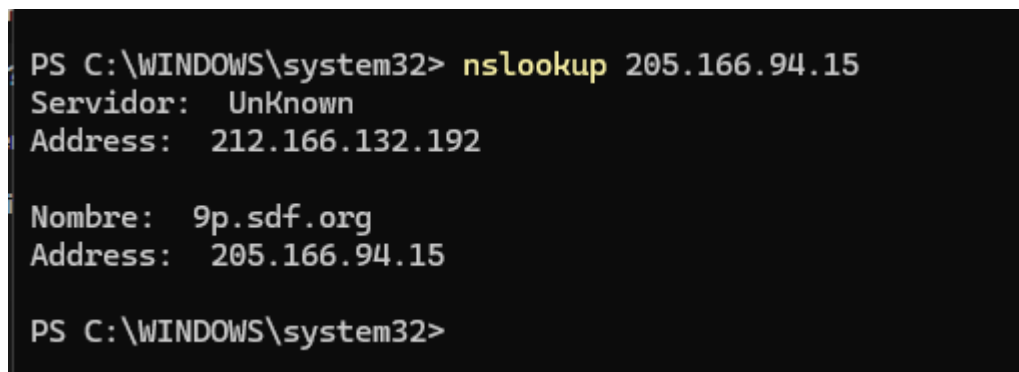
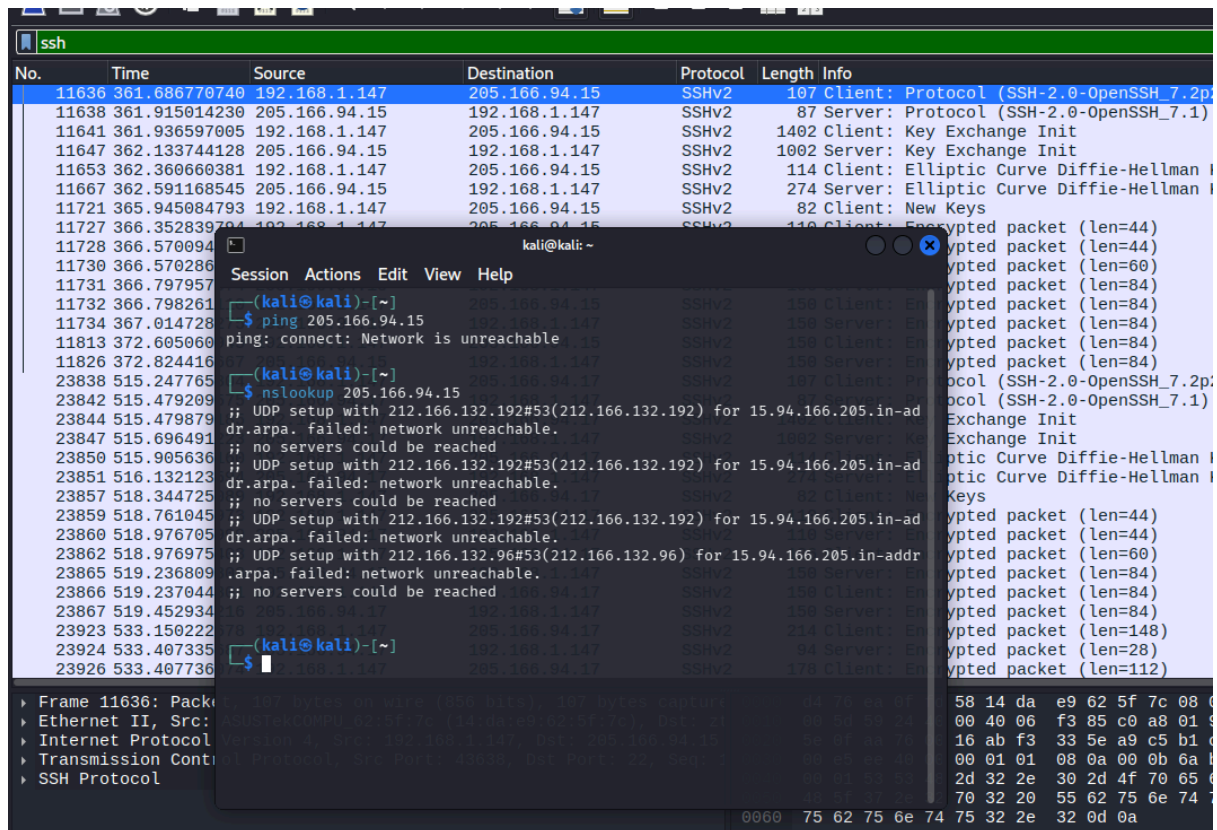
9.

No.	Time	Source	Destination	Protocol	Length	Info
27367	924.863320940	192.168.1.147	94.142.241.111	TELNET	93	Do Suppress Go Ahead, Will Terminate
27368	924.956222949	94.142.241.111	192.168.1.147	TELNET	72	6 bytes data
27370	925.046956411	94.142.241.111	192.168.1.147	TELNET	1054	988 bytes data
27390	931.341644664	192.168.1.147	94.142.241.111	TELNET	68	2 bytes data
27392	931.629083995	94.142.241.111	192.168.1.147	TELNET	1054	988 bytes data
27394	934.094543546	192.168.1.147	94.142.241.111	TELNET	68	2 bytes data
27428	941.644671357	94.142.241.111	192.168.1.147	TELNET	1054	988 bytes data
27430	941.787108791	94.142.241.111	192.168.1.147	TELNET	1054	988 bytes data
27440	942.803837926	94.142.241.111	192.168.1.147	TELNET	1054	988 bytes data



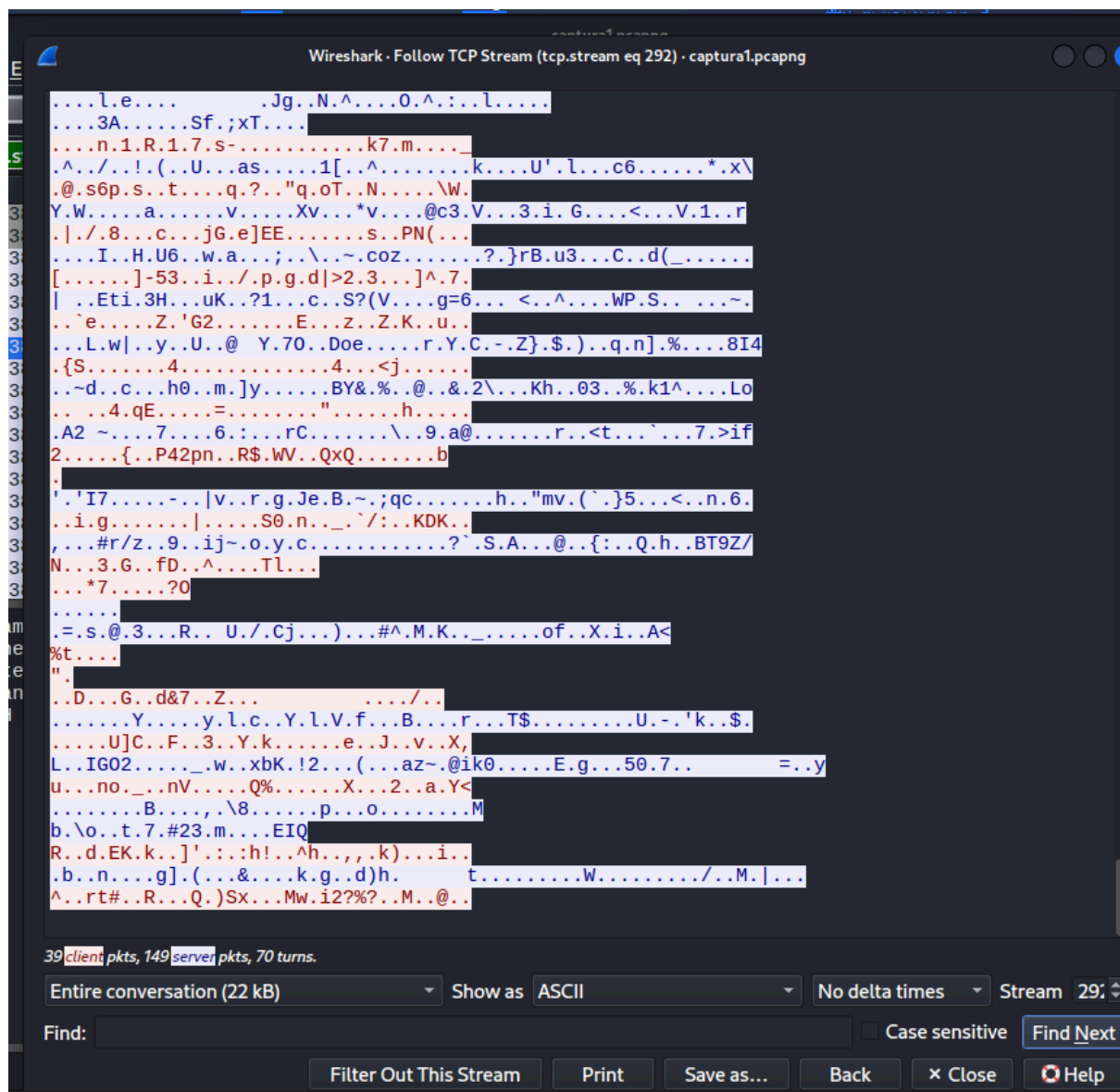
A la captura de Wireshark podem observar que la IP de destí del servidor Telnet és la 94.142.241.111. Per esbrinar a quin domini pertany aquesta adreça, he obert la terminal de Kali Linux i he executat la comanda nslookup 94.142.241.111. El resultat d'aquesta comanda ens indica que la IP pertany al domini towel.blinkenlights.nl."

10.



A la captura de Wireshark podem observar com el nostre equip es connecta a un servidor SSH extern amb la IP 205.166.94.15. En intentar esbrinar a quin domini pertany executant la comanda nslookup 205.166.94.15 des de la màquina virtual Kali, m'ha donat un error de connexió ('network is unreachable', com es pot veure a la primera captura adjunta). Per poder completar l'exercici, he hagut d'executar la mateixa comanda des del meu ordinador físic. El resultat final demostra que aquesta adreça IP correspon al domini 9p.sdf.org.

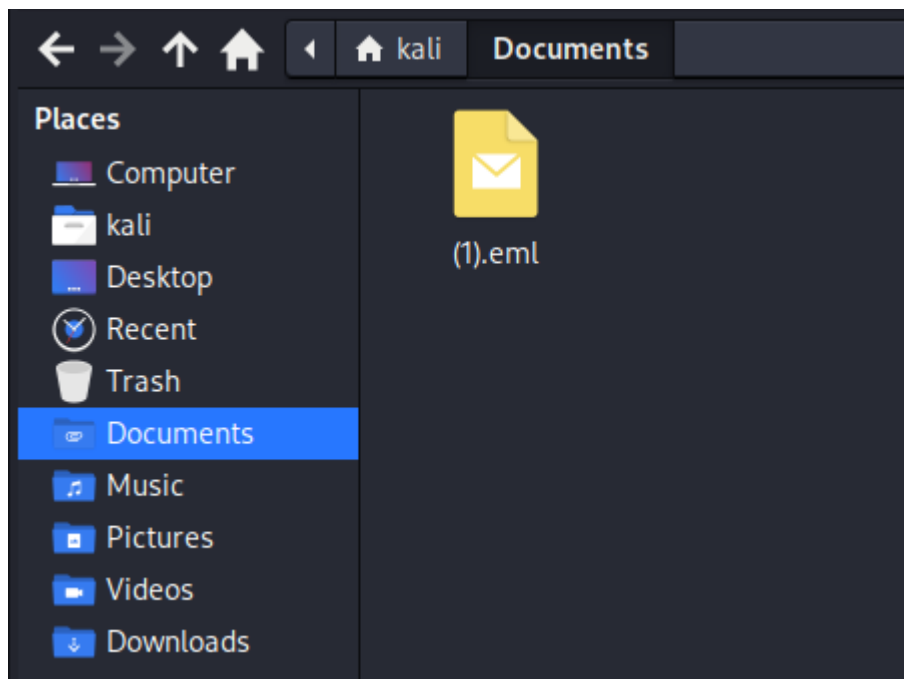
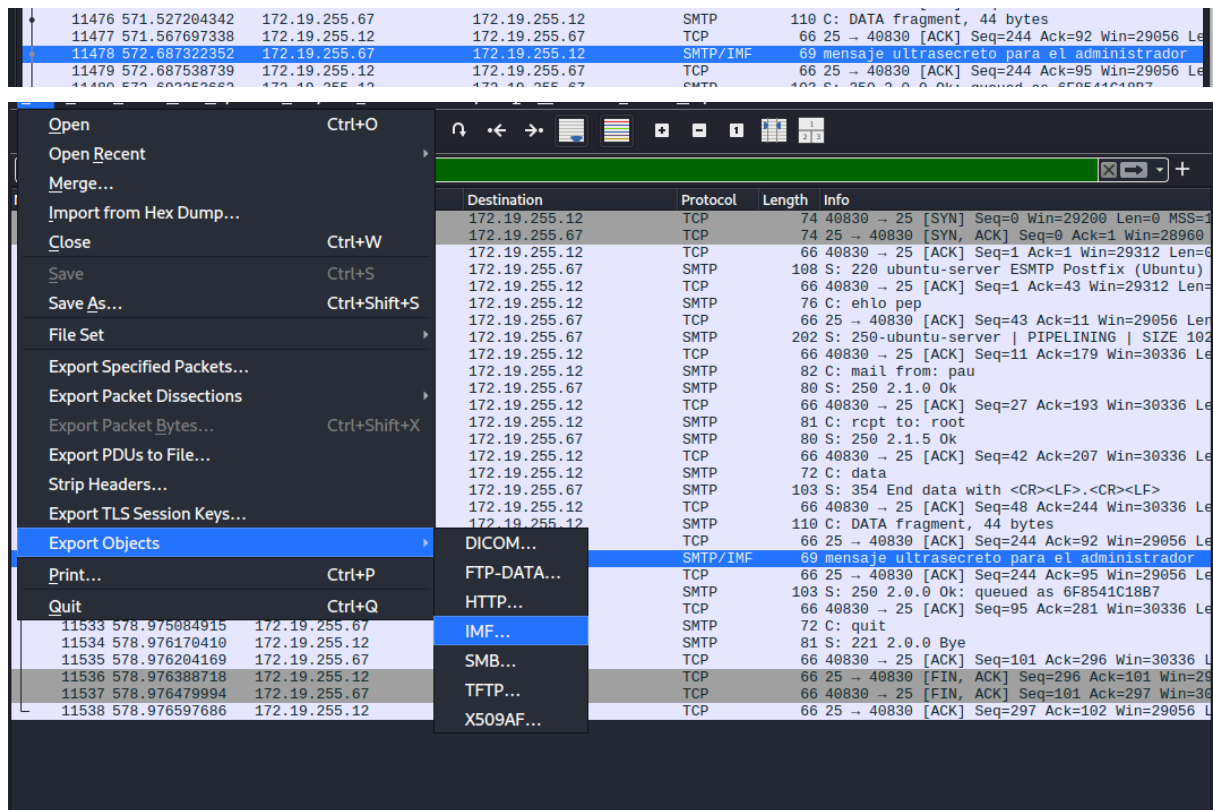
11.



No, no podem llegir el que s'està enviant. Això passa perquè el protocol SSH és segur i xifra (amaga) tota la informació perquè ningú la pugui espiar. En lloc de veure el text normal, només veiem símbols estranys i lletres sense sentit, com per exemple aquest trosset del paquet:

.....l.e.....Jg..N.^.....O.^:..l.....3A.....Sf;xT.....n.1.R.1.7.s-.....k7.m....._

12.



Per extreure el fitxer enviat a través de correu sortint (SMTP), he anat al menú superior de Wireshark i he utilitzat l'eina d'extracció automàtica: **File -> Export Objects -> IMF**. Aquesta opció analitza els paquets amb format IMF (Internet Message Format), com el paquet 11478 de la captura, i permet guardar el contingut o fitxer adjunt del correu directament a l'equip, tal com es mostra a la captura de pantalla adjunta.